OBD2 sniffer: revision A

A read-only CAN and OBD2 scanner. Powered from the laptop's USB port only.
Design notes and requirements: docs/design/circuit-board.md, docs/design/requirements.md.
Parts, datasheets and the pre-order checklist: hardware/rev-a/README.md.

Sheets

USB and power



File: usb-power.kicad_sch

USB-C socket, fuse, ESD protection,
3.3 V regulator, PWR light.

Microcontroller



File: microcontroller.kicad_sch

STM32F446RET6, crystal, decoupling, reset and boot,
SWD and debug serial headers, BUS light.

Sheet: /
File: obd2-sniffer-rev-a.kicad_sch

Title: OBD2 sniffer, revision A

Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/3

USB and power

The board is powered from the laptop's USB port only (PWR-5).
The car's 12 V supply has no connection to the board (PWR-8).

USB-C socket

5.1 k on each CC pin: the host switches its 5 V on.

Fuse

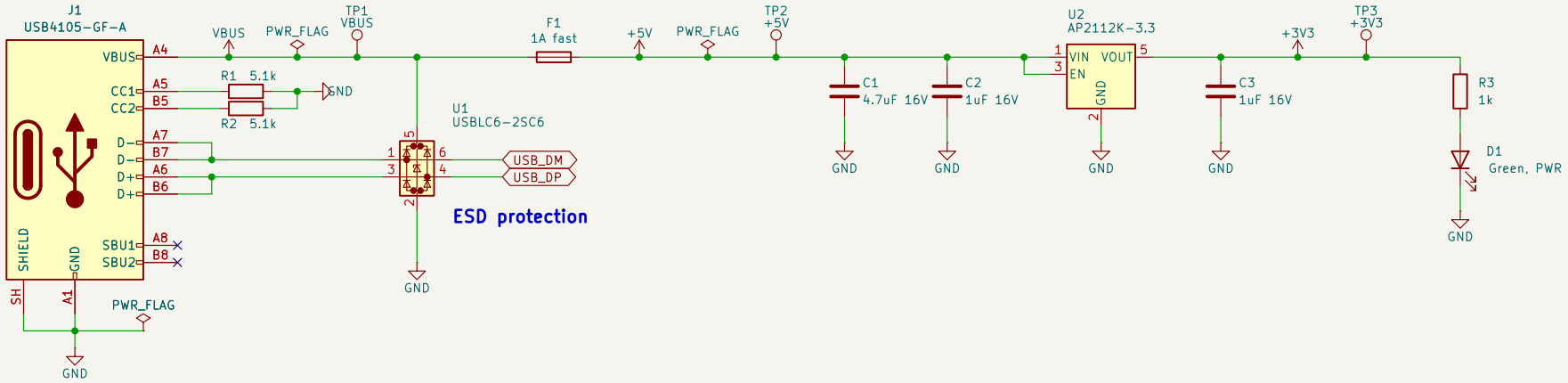
1 A fast, 0.13 ohm: a short on the board can't draw a dangerous current.

3.3 V regulator

1 uF ceramic in and out, as the datasheet asks. Enable is tied to the input: always on.

PWR light

On whenever 3.3 V is present, with or without firmware (USE-3).



The socket's shield is tied to ground. SBU pins are not used.

Total capacitance on the USB 5 V stays under 10 uF, the limit a USB device may present when plugged in.
The transceivers take the 5 V directly and need at least 4.5 V: the fuse drops about 35 mV at 250 mA.

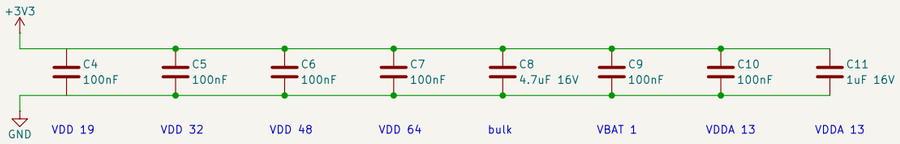
Sheet: /USB and power/		
File: usb-power.kicad_sch		
Title: USB and power		
Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/3

Microcontroller

STM32F446RET6 in LQFP64, the same chip as the development board.
Supply pins, capacitors and boot pins follow ST's datasheet DS10693 and application notes AN4488 and AN2606.

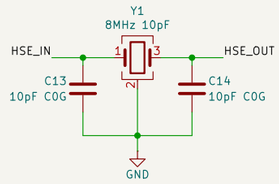
Decoupling

100 nF at each supply pin, placed at the pin. One 4.7 uF for the chip.
VDDA joins 3.3 V directly: the analogue converter isn't used.



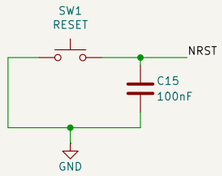
8 MHz crystal

ECS ECX-53B, 10 pF load, at most 100 ohm.
Load capacitors: 2 x (10 pF − 5 pF stray) = 10 pF.
USB and the USB bootloader need this clock.



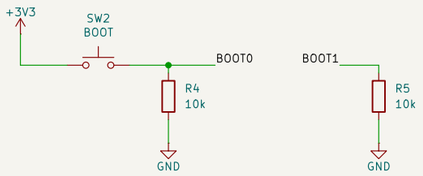
Reset

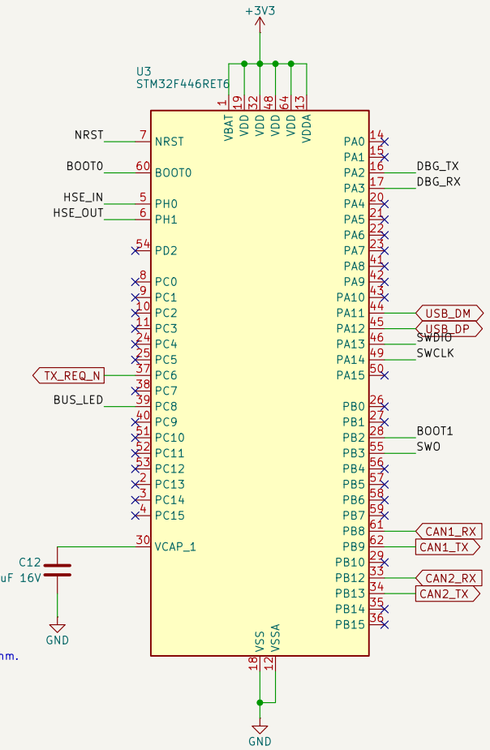
The chip has its own pull-up on NRST.



Boot

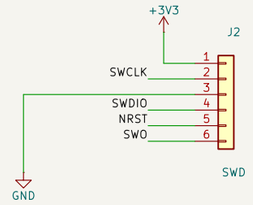
Hold BOOT while pressing RESET to start the chip's USB bootloader (USE-4).
BOOT0 is low otherwise, and BOOT1 (PB2) is always low, as the bootloader needs.





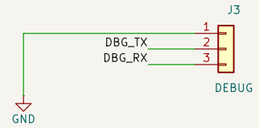
SWD debug header

For programming and debugging.
Pin 1 lets the debugger sense 3.3 V.



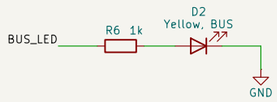
Debug serial header

Start-up and crash reports, 3.3 V levels.



BUS light

Driven by firmware (USE-3).



Sheet: /Microcontroller/		
File: microcontroller.kicad_sch		
Title: Microcontroller		
Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/3